A Research-Informed, Explainable and Context-Aware Travel Recommendation Prototype for Maharashtra

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Abstract

Travel recommendation systems must balance personal preferences with practical constraints, contextual conditions, and the need to show users why a suggestion is appropriate. Recent research identifies persistent difficulties including sparse or outdated interaction data, weak contextual adaptation, poor diversity, and limited representation of individual travel needs. This manuscript documents a research-informed revision of TripGenie, a Maharashtra-first travel discovery and planning prototype built with Flask, SQLite, CSV data, and a lightweight K-nearest-neighbours (KNN)-style profile similarity component. A focused review of recent tourism-recommender literature is mapped to five practical modifications: factor-level explanation of each recommendation score, category-aware shortlist diversification, a per-user explicit feedback loop, forecast-aware ranking for near-term travel dates, and a cold-start fallback for destinations absent from the training labels. Six functional unit tests cover key ranking behaviours. The system continues to use synthetic training examples and illustrative hotel data; therefore, the implementation makes no claim of real-world prediction accuracy or improved traveller satisfaction. The work is a reproducible engineering prototype and gap-analysis artefact. Future evaluation requires verified destination data, real or appropriately licensed anonymized travel interactions, baseline comparisons, and a user study.

Keywords: tourism recommender system; explainable recommendation; context-aware ranking; explicit feedback; travel planning; Maharashtra; cold start; recommendation diversity

1. Introduction

Travel planning combines several connected decisions: where to go, which experiences fit a traveller's interests, how much time and money a trip may require, how to move between locations, and what conditions may affect the visit. A single popularity score does not explain these trade-offs. Travel is also context-dependent: a destination that suits a traveller in one season or weather condition may be less suitable in another. Recent work argues that travel recommender systems should be treated as decision-support tools that combine contextual information, diverse recommendations, interaction data, and transparent trade-offs rather than as simple next-item predictors [1].

TripGenie is a student project oriented first towards Maharashtra and Indian destinations. It covers forts, caves, engineering and geological wonders, wildlife attractions, nature escapes, and general travel destinations. Its current prototype includes a Flask web application, SQLite storage, CSV destination/place/hotel data, a KNN-style similarity artefact, interactive maps, weather display, itinerary pages, and outbound navigation/transport links. Earlier versions returned a shortlist whose factor contributions were not visible and that did not learn from an individual's explicit reaction to the suggestions. The training profiles and demo hotel records are synthetic/illustrative, which creates a clear risk if they are mistaken for measured travel behaviour or live commercial data.

This manuscript has two purposes: (1) connect recent literature to specific gaps in the project and (2) document a modest implementation intended to address feasible gaps without overstating what the available data can prove.

2. Focused literature review

This is a focused review of selected papers, not a systematic review. Sources were selected for direct relevance to contextual tourism recommendation, route feasibility, data sparsity, diversity, feedback, and personalized travel planning.

2.1 Dynamic context and multiple objectives

Qassimi and Rakrak introduce a multi-objective contextual multi-armed bandit approach for smart tourism. Their framing highlights the challenge of incorporating time, weather, location, changing preferences, and more than one objective; their method learns from feedback and examines relevance alongside fairness [2]. For a small project, implementing a full contextual-bandit policy would require interaction volume and evaluation infrastructure that TripGenie does not yet have. The practical first step is a transparent multi-factor score and a controlled feedback signal.

2.2 Heritage constraints and spatiotemporal route planning

Yan, Wang and Lu discuss personalized tourism recommendation and route optimization in cultural-heritage contexts, including spatial and temporal dynamics and site-specific constraints. The paper also acknowledges limitations tied to static historical data and representativeness [3]. TripGenie can account for duration and approximate proximity, but it does not yet verify live opening hours, closures, visitor capacity, trek conditions, accessibility, or conservation restrictions. Those constraints remain important future work rather than features the prototype should imply it already solves.

2.3 Context, diversity and sparse interaction data

Kamal and Chatzigiannakis present a touristic-experience recommender that incorporates contextual signals such as weather and traffic, and report experiments on several tourism-platform datasets. They identify data availability, dynamicity, incompleteness, dataset bias and limited user aspects as limitations [4]. Xiao et al. model tourism trajectories with temporal and sequential information using a large trajectory collection from Chengdu [5]. These papers illustrate the gap between research-scale trajectory modelling and TripGenie's limited synthetic profile file. TripGenie therefore does not attempt to reproduce a deep sequential model; it introduces date-aware forecast adjustment, cold-start metadata handling and explicit feedback while documenting that better real data are needed.

2.4 Personalized travel planning and a very recent research direction

Singh et al. investigate personal LLM agents for tailored travel planning and report that personal user models can improve plan relevance compared with generic plans in their evaluation [6]. Bellogín et al.'s 6 October 2026 preprint argues that travel recommenders still face sparse/outdated datasets, overemphasis on prediction accuracy relative to novelty/contextual relevance, and a failure to address specific traveller needs; it calls for flexible, transparent decision support combining recommendation techniques [1]. The latter is a preprint and should not be treated as peer-reviewed evidence. It is useful as a very recent statement of open problems, not as proof that one implementation is superior.

Esper et al. study itinerary recommendation alongside mobile edge-resource allocation and note that advanced digital experiences add application-quality and resource constraints to travel planning [7]. Such infrastructure-aware optimization is outside the scope of TripGenie's current college demo, but it reinforces that route relevance alone is not the whole travel experience.

3. Gap analysis and implementation mapping

Research gap	Evidence from literature	TripGenie response in this revision	Remaining limitation
Weak explanations for recommendations	Context-aware and modern systems must make trade-offs legible; travel decisions involve money/time and user trust [1, 2].	Each card shows factor-level scores and their configured weights.	Weights are hand-set and need user validation; explanations are not causal explanations of a trained deep model.
Static recommendations that do not learn from user reactions	Contextual-bandit work emphasises feedback and adaptation [2]; personal user models are studied in travel planning [6].	Logged-in users can choose “More like this” or “Not for me”; the signal is stored in SQLite and changes later rankings.	Feedback is sparse, explicit, and not a full online-learning or contextual-bandit implementation.
Repetitive recommendations and weak coverage	Recent research highlights diversity, item coverage and novelty [1, 4].	The returned shortlist limits repetition of a single category where alternatives exist.	The heuristic can still under-represent niche interests and requires measured diversity evaluation.
Inadequate weather/date context	Context-aware tourism research identifies time, weather and location as relevant inputs [2, 4].	The user selects a travel date; available forecasts within the near-term forecast window can adjust a weather-fit factor.	A failed/unavailable forecast is neutral. Forecast is advisory, not a closure or safety check; dates outside the forecast window are not weather-ranked.
Cold-start destinations have no training examples	Travel recommenders face sparse interaction data and new-user/new-item cases [1, 4].	Destinations still receive scores from metadata factors; unseen training labels receive a modest neutral prior rather than a zero similarity.	Synthetic training data still cannot establish real preference patterns.
Incomplete, stale or unrepresentative data	Data sparsity, age, bias and incompleteness are explicit concerns [1, 3, 4].	A data-quality register documents which files are synthetic or illustrative and lists source verification as a production requirement.	A full official-source inventory with source_url and last_verified for every destination/place has not yet been completed.
Itinerary feasibility and on-site constraints	Heritage routes may need opening-hour, capacity and conservation constraints [3].	Current ranking includes trip duration and straight-line proximity; the website retains maps/routes and external directions.	It does not validate current operating hours, site restrictions, route accessibility or total live journey costs.

4. Prototype method

4.1 Inputs and data

The user submits a total budget, trip duration, number of travellers, interest, origin city and intended travel date. Budget is converted into an approximate per-person-per-day value to compare with destination metadata. Destination records include name, location/state, category, interest tags, expected daily budget, ideal duration, coordinates, description and image URL. The current package has 38 destination records, 128 place records, 114 illustrative hotel rows, and 2,067 synthetic training profiles. These figures describe the included demo files, not verified real-world coverage.

4.2 Explainable ranking

Each destination receives factor scores on a 0–100 scale. The configured weighted score is:

$$S = 0.28I + 0.23B + 0.12D + 0.12K + 0.10P + 0.10W + 0.05F$$

where I is interest match, B is approximate budget fit, D is duration fit, K is training-profile similarity, P is proximity fit, W is forecast fit, and F is prior user-feedback fit. The seven weights sum to 1.00. Final display values are clipped to a 1–99 range to avoid presenting them as calibrated probabilities. The factor

breakdown is rendered directly on each recommendation card. Weights are engineering defaults, not values optimized through a training or validation procedure.

Budget fit is an approximate comparison against a destination-level daily estimate, not a quote for transport, hotels, food and activities. Proximity uses Haversine straight-line distance between approximate city/destination coordinates, not road distance. Training-profile similarity is derived from the KNN-style matching component over synthetic examples; it must not be interpreted as evidence that the user resembles actual historical travellers.

4.3 Forecast context and graceful fallback

The selected date is checked against the near-term forecast window. When the date falls within the next seven days, the engine makes a best-effort request to Open-Meteo for daily weather code and precipitation probability for a limited set of leading candidates. Results are cached for 15 minutes in the process. For outdoor-tagged interests, high predicted precipitation or severe-weather codes reduce the weather-fit factor and produce a note to consider a backup plan. If a request fails, a coordinate is unavailable, or the chosen date lies outside the supported window, the engine uses a neutral weather factor and states that a forecast was not used. This mechanism is advisory and does not replace official closure announcements or local safety advice.

4.4 Explicit feedback and diversification

A logged-in user can mark a recommendation as positive or negative. The application stores user ID, destination, signal, selected interest, travel date and timestamp in a separate SQLite table. On a subsequent request, prior signals for that user influence the feedback factor; when a specific interest is selected, feedback is filtered to that interest. The initial signal is intentionally low-weight (5%) because a few clicks do not provide enough evidence for a strong learned policy.

The final shortlist is diversified by category, limiting any one category to three items where alternatives exist. This is a transparent re-ranking heuristic, not a novelty model or fairness guarantee. Cold-start destinations absent from the training file still receive scores from metadata; the model similarity factor falls back to a modest prior rather than treating them as impossible recommendations.

5. Implementation and functional verification

The revision changes recommendation_engine.py to expose factor scores, incorporate a near-term weather signal when available, use accumulated user feedback, and diversify the shortlist. app.py adds the feedback table, handles a travel date from the form, retrieves per-user feedback, and stores new feedback posts. The recommendation template presents factor bars, weather notes and feedback controls. DATA_QUALITY_AND_RESEARCH_GAPS.md records unresolved data-quality and validation limits.

Six unit tests passed in the project working environment. They check that factor explanations are returned; interest fit distinguishes relevant and unrelated candidates; positive feedback changes the feedback factor and score; a mocked forecast changes weather fit; dates outside the forecast window do not trigger a weather request; and category repetition is limited where alternatives exist. These are functional unit tests. They do not measure precision, recall, NDCG, user satisfaction, factual accuracy, forecast correctness, fairness or performance under production load. The complete Flask server/browser journey should still be exercised in the developer's local environment.

Reproduction commands from the project root:

```
py -m pip install -r requirements.txt
py train_model.py
py -m unittest discover -s tests -v
py app.py
```

6. Limitations and responsible use

The most important limitation is data validity. The 2,067 training profiles are synthetic and were created for a demonstration. The demo hotel rows and prices are illustrative, not live or bookable. Destination coordinates and descriptions are a curated starter dataset, not an exhaustive or continuously verified source. The application must not claim validated recommendation accuracy, real-time hotel availability, guaranteed trip budgets, or safe access to sites that may be closed or affected by weather.

A stronger research evaluation should use public or permissioned/anonymized travel interactions, document source provenance for each place, and compare the proposed hybrid ranker against clear baselines (metadata-only, KNN-profile-only and the prior version). Appropriate offline metrics include Recall@K or NDCG@K for relevance, catalogue coverage and category diversity for variety, constraint-violation counts for feasibility, and latency for practical use. A user study would be needed to measure perceived explanation usefulness, trust, and satisfaction. Separate checks are also required for demographic/accessibility coverage and sustainability effects before making claims about fairness or responsible tourism.

Other open gaps include verifying official opening hours and closures; public transport route feasibility; accessibility, trek difficulty and conservation constraints; real hotel rates and transport fares; group travel preference aggregation; privacy controls for feedback history; and conversational/LLM planning grounded in reliable source content. These are future directions, not features claimed as complete in this revision.

7. Conclusion

Recent travel-recommender literature motivates systems that account for context, user feedback, diversity, personal needs, and data quality. TripGenie now implements a modest subset of these recommendations through explainable multi-factor scores, a date-aware forecast factor with a neutral fallback, category diversification, per-user explicit feedback, and cold-start handling. Six unit tests verify core behaviour, but no real-world accuracy or user-satisfaction claims are justified by synthetic data. The most valuable next step is not simply a more complex neural model: it is to build a provenance-tracked destination dataset, verify travel constraints, secure real evaluation data, and measure the enhanced system against baselines.

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